SentinelOps

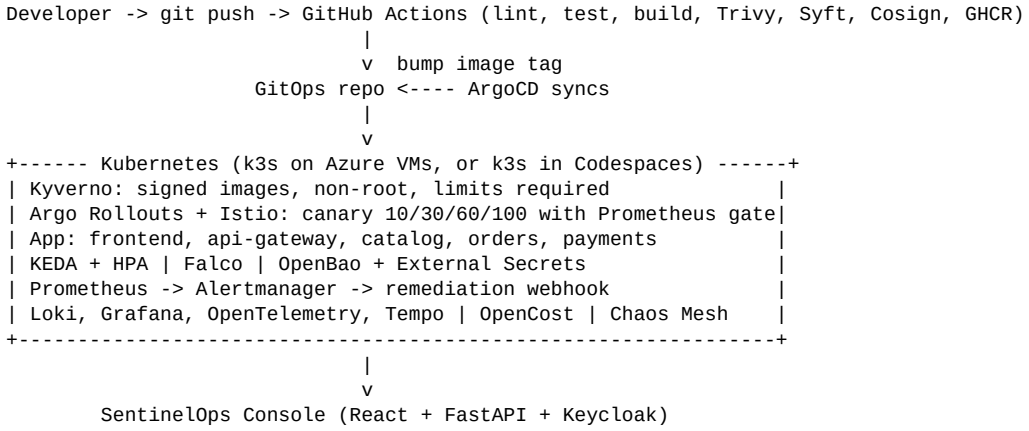
Ship fast. Stay secure. Self-heal. SentinelOps is a self-healing, cost-aware, secure Kubernetes platform. A developer pushes code to Git; the platform builds it, scans it, signs it, deploys it as a canary, watches it, rolls it back if it misbehaves, scales it, recovers it from failures, and reports what it costs. A custom web console ties it together.

> **Status:** Planning. The stack matches the parent blueprint (Advanced Cloud Computing Project Ideas, Part 2). Only hosting changed: **no credit card and nothing local**. Versions are mid-2026 known-good picks; check release pages and pin the latest stable patch before starting.

1. Problems Solved

Real-world problem	What SentinelOps does	Target result
Slow, manual deployments	GitOps and automated CI/CD	Commit to production under 5 min
Vulnerable code reaches production	Trivy, Kyverno, image signing, SBOM	100% of insecure test deployments blocked
Bad releases cause outages	Canary releases with automatic rollback	Rollback under 60 s
Failures need a human at 3 a.m.	Self-healing, alert-driven remediation	Recovery under 30 s
Idle workloads waste money	KEDA scale-to-zero, right-sizing, OpenCost	Cost reduction of 30%+

2. Architecture



3. Tech Stack

Area	Tools and versions
Infrastructure	Ubuntu Server 24.04 LTS, OpenTofu 1.10.x, Ansible core 2.18.x, k3s v1.33.x, Helm 3.18.x, Kustomize 5.6.x
CI/CD and GitOps	GitHub Actions, Docker Buildx 0.24.x, GHCR, ArgoCD 3.0.x or later 3.x, Argo Rollouts 1.8.x, Argo Image Updater 0.16.x (optional)

Area	Tools and versions
Security	Trivy 0.6x.x, Trivy Operator 0.2x.x, Syft 1.x, Cosign 2.5.x, Kyverno 1.14.x, Falco 0.41.x, OpenBao 2.x, External Secrets Operator 0.18.x, cert-manager 1.18.x, Gitleaks 8.2x.x
Observability	kube-prometheus-stack 7x.x (Prometheus 3.x, Alertmanager 0.28.x), Grafana 12.x, Loki and Promtail or Alloy 3.5.x, Tempo 2.8.x, OpenTelemetry Collector 0.12x.x
Scaling, cost, chaos, mesh	KEDA 2.17.x, metrics-server 0.7.x, Goldilocks 4.x and VPA 1.4.x, OpenCost 1.11x.x, Chaos Mesh 2.7.x, k6 1.x, Istio 1.26.x
Demo app	Go 1.24, Python 3.12 and FastAPI 0.115, React 19, Vite 6, TypeScript 5.8, PostgreSQL 17 with CloudNativePG 1.26, Valkey 8
Console	React 19, Vite 6, TypeScript 5.8, Tailwind CSS 4, shadcn/ui, Recharts 2.15, TanStack Query 5; FastAPI 0.115, httpx, Pydantic 2.11, kubernetes client 33; Keycloak 26
Testing	pytest, Go test, Playwright 1.5x, k6

4. Free, No-Card, Cloud-Only Hosting

Track	Who	What you get
A. Azure for Students	Verified university students	$100 credit for 12 months, renewable yearly; no credit card; two Arm VMs (control plane and worker) provisioned by OpenTofu and configured by Ansible
B. GitHub Codespaces	Anyone with a GitHub account	120 core-hours and 15 GB storage per month on the free plan; no card (usage is blocked, not billed, when the quota ends); k3s runs inside Docker using k3d

Run `make down` after each session on Track A to deallocate VMs and save credit. Re-check both offers' current terms before you start.

5. Repositories

```
sentinelops-app/       services/ (api-gateway, catalog, orders, payments, frontend), CI
sentinelops-gitops/    apps/, platform/, workloads/, policies/, rollouts/, chaos/
sentinelops-infra/     tofu/ (Azure VNet, NSG, VMs), ansible/, Makefile
sentinelops-console/   frontend/ (React), backend/ (FastAPI)
```

6. Quick Start (Track A)

```
git clone https://github.com/<you>/sentinelops-infra.git && cd sentinelops-infra
az login                    # student subscription, no card
make up                     # OpenTofu creates network + VMs, Ansible installs k3s
helm install argocd ...     # bootstrap ArgoCD once
kubectl apply -f root-app.yaml   # app-of-apps installs everything else
make demo                   # runs the 3-minute demo scenarios
make down                   # deallocate VMs to save credit
```

7. Console Screens

Login, Platform Overview, Deployments and Releases, Security Center, Incidents and Self-Healing, Cost and Capacity, Chaos Lab, Audit Log. Styling lives in `global.css` (Tailwind 4 theme, shadcn/ui tokens, buttons, gradients, dark mode).

8. Success Metrics

Metric	Target
Commit to running in prod	under 5 min
Insecure deployments blocked	100%
Automatic rollback time	under 60 s
Pod failure recovery	under 30 s, error rate under 1%
Cost reduction (autoscaling on vs fixed)	30% or better
Availability under chaos	99.5% or better

9. Roadmap (10 weeks)

Weeks	Focus
1-2	Repos, OpenTofu and Ansible, k3s, ArgoCD, demo microservices
3-4	CI with Trivy, Syft, Cosign; Kyverno, OpenBao, Istio mTLS
5-6	Observability stack; canary releases; HPA and KEDA
7-8	Remediation webhook, Falco, Chaos Mesh, OpenCost, load tests
9-10	Console and Keycloak, docs, demo video, blog post

10. License

Proposed: Apache-2.0. All platform components are open source.